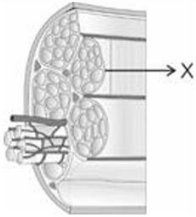


Locomotion and Movement

1. Given below is a diagrammatic representation of the cross sectional view of a muscle. Choose the option that best represents the structure marked as 'X'.



- (1) Fascicle
(2) Epimysium
(3) Endomysium
(4) Myofilament
2. Calcium is important in skeletal muscle contraction because it
- (1) binds to troponin to remove the masking of active sites on actin for myosin.
(2) activates the myosin ATPase by binding to it.
(3) prevents the formation of bonds between the myosin cross bridges and the actin filament.
(4) detaches the myosin head from the actin filament.
3. Mechanism of muscle contraction is best explained by.
- (1) Physical filament theory
(2) Chemical filament theory
(3) Sliding filament theory
(4) Jumping filament theory
4. Thin myofilaments do not consist of
- (1) Myosin
(2) Actin
(3) Tropomyosin
(4) Troponin
5. Statement-I: Repeated activation of the muscles can lead to the accumulation of lactic acid due to anaerobic breakdown of glycogen in them, causing fatigue.
Statement-II: Muscle contains a red coloured oxygen storing pigment called myoglobin.
- (1) Statement I and Statement II both are correct.
(2) Statement I is correct, but statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Statement I and Statement II both are incorrect.
6. Assertion (A): The 8th, 9th and 10th pairs of ribs are called vertebrochondral (false) ribs.
Reason (R): These ribs do not articulate directly with the sternum but join the seventh rib with the help of hyaline cartilage.
- In light of above statements, choose the correct option.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false
7. What would happen if ATP suddenly were not available after the sarcomere had started to shorten?
- (1) Cross bridges would not be able to detach from actin
(2) Muscle would remain in a state of rigidity
(3) Muscle would relax immediately
(4) Both (1) & (2)

8. Match the column

Column-A		Column-B	
(A)	Structural and functional unit of myofibril	(i)	H zone
(B)	Protein of thin filament	(ii)	Myosin
(C)	Protein of thick filament	(iii)	Sarcomere
(D)	The central part of thick filament not overlapped by thin filament.	(iv)	Actin

- (1) A-i; B-ii; C-iii; D-iv
 (2) A-i; B-iii; C-ii; D-iv
 (3) A-i; B-iv; C-iii; D-ii
 (4) A-iii; B-iv; C-ii; D-i

9. Which of the following events occur(s) during muscle contraction in skeletal muscle?

- (i) 'I' band reduces in length
 (ii) 'A' band reduces in length
 (iii) 'H' zone greatly reduces in size

Choose the correct option.

- (1) (i) and (ii)
 (2) (ii) and (iii)
 (3) (i) and (iii)
 (4) Only (i)

10. Match the columns and choose the correct option

Column-A		Column-B	
(A)	Balls socket joint	(i)	between humerus and pectoral girdle
(B)	Hinge joint	(ii)	between carpals and metacarpals
(C)	Gliding joint	(iii)	in between the carpals
(D)	Saddle joint	(iv)	Knee joint

- (1) A-1, B-4, C-3, D-2
 (2) A-1, B-4, C-2, D-3
 (3) A-2, B-3, C-4, D-1
 (4) A-4, B-1, C-3, D-2

11. Identify the incorrect statement w.r.t. scapula

- ((1) It is a large triangular flat bone.
 (2) It is situated in the ventral part of the thorax between 3rd and 8 th ribs.
 (3) The dorsal, flat, triangular body of scapula has a slightly elevated ridge called the spine.
 (4) Spine projects as a flat, expanded acromion process.

12. The ____A____ releasing the ADP and Pi goes back to its ____B____ state. A new ATP binds and the cross bridge is ____C____. The ATP is again hydrolysed by the ____D____ head and cycle of cross bridge formation and breakage is repeated causing further sliding.

Fill the appropriate option in the blanks A, B, C and D.

- (1) A-myosin, B-relaxed, C-broken, D myosin
 (2) A-actin, B-relaxed, C-broken, D myosin
 (3) A-myosin, B-relaxed, C-broken, D actin
 (4) A-actin, B-relaxed, C-broken, D actin

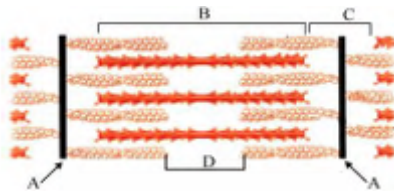
13. Assertion (A): Ribs have two articulation surfaces. Reason (R): Each rib is a thin flat bone connected dorsally to the sternum and most of them are connected ventrally to the vertebral column.

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
 (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
 (3) Assertion (A) is true, and Reason (R) is false.
 (4) Assertion (A) is false, and Reason (R) is true.

14. Total number of skull bones is

- (1) 20
 (2) 22
 (3) 21
 (4) 23

15. Z line shown in this diagram is



- (1) D
- (2) C
- (3) B
- (4) A

16. **Assertion(A):** Cardiac muscles are involuntary in nature.

Reason(R): The nervous system control their activities directly.

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

17. **Complete the analogy w.r.t the type of joints present in humans. Ball and socket joint : Between humerus and pectoral girdle :: Saddle joint : Between _____** Choose the correct option.

- (1) Carpal and metacarpal of thumb
- (2) The carpals
- (3) Adjacent vertebrae
- (4) Flat skull bones

18. **Statement-I:** Progressive degeneration of skeletal muscles mostly due to genetic disorder muscular dystrophy.

Statement-II: Tetany is a age related disorder

- (1) I and II both are correct.
- (2) I is correct, but II is incorrect.
- (3) I is incorrect, but II is correct.
- (4) I and II both are incorrect.

19. **Statement-I:** Below the acromion is a depression called the glenoid cavity which articulates with the head of the humerus to form the shoulder joint.

Statement-II: The two halves of the pelvic girdle meet dorsally to form the pubic symphysis containing fibrous cartilage.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

20. **Which of the following is a flat bone one the ventral midline of thorax?**

- (1) Sternum
- (2) Ribs
- (3) Thoracic vertebrae
- (4) Atlas

21. **How many statements are false?**

- A. Suture joint is a type of fibrous joint between cranial bones
- B. Myasthenia gravis is an autoimmune disorder
- C. Tetany results in rapid spasm or wild contraction in nerve.
- D. Accumulation of uric acid crystals in joints occur in gout.

- (1) One
- (2) Two
- (3) Three
- (4) Four

22. **Gout is the inflammation of joints due to accumulation of;**

- (1) Urea crystal
- (2) NH_3
- (3) Uric acid crystal
- (4) CaCO_3 crystals

23. Assertion (A): Osteoporosis is characterised by increased chances of fractures.

Reason (R): Decreased levels of estrogen is a common cause of osteoporosis.

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

24. Which of the following pair is correctly matched?

- (1) Hinge joint – Between the vertebrae
- (2) Gliding joint – Between the carpals
- (3) Cartilaginous joint – Between atlas and axis
- (4) Fibrous joint – Between carpal and metacarpal of thumb

25. Assertion : Human has dicondylic skull

Reason : Skull articulates with superior region of the vertebral column with the help of two occipital condyles.

- (1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) If Assertion is true but Reason is false
- (4) If both Assertion and Reason are false

26. Assertion : First 7 pairs ribs are called true ribs.

Reason : These ribs are not connected ventrally to sternum

- (1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) both Assertion & Reason are true but reason is not the correct explanation of the assertion
- (3) If Assertion is true but Reason is false
- (4) If both Assertion and Reason are false

27. Each myofibril contains many serially arranged contractile units called

- (1) Muscle fibre
- (2) Sarcomere
- (3) Myofilament
- (4) Muscle bundle

28. Which of the following statement is correct regarding joints?

- (1) Joints are essential for all types of movements involving the bony parts of the body.
- (2) Joints are points of contact between bones, or between bones and cartilages.
- (3) Force generated by the muscles is used to carry out movement through joints.
- (4) All of these

29. Statement-I: Axial skeleton comprises 126 bones distributed along the main axis of the body.

Statement-II: The skull region articulates with the superior region of the vertebral column with the help of three occipital condyles.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

30. Which of the following is a part of pectoral girdle?

- (1) Ilium
- (2) Ischium
- (3) Acetabulum
- (4) Glenoid cavity

31. Ilium is a part of

- (1) Hindlimbs
- (2) Pelvic girdle
- (3) Pectoral girdle
- (4) Forelimbs

32. Assertion (A): Carpals are the palm bones.

Reason (R): Wrist bones are 8 in number.

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

33. Assertion : Bone has calcium salts in its matrix whereas cartilage pliable matrix.

Reason : Bone has calcium salts in its matrix whereas cartilage has chondroitin salts in its matrix

- (1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) If Assertion is true but Reason is false
- (4) If both Assertion and Reason are false

34. In osteoporosis bone mass.....which the chance of fracture.....

- (1) Increases, decreases
- (2) Increases, increases
- (3) Decreases, increases
- (4) Decreases, decreases

35. Which of the following statement is true about cartilaginous joints?

- (1) Do not allow any movement.
- (2) Characterised by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones.
- (3) Allows considerable movement.
- (4) These joints are present between the adjacent vertebrae in the vertebral column.

36. (i) Auto immune disorder affecting

neuromuscular junction leading to fatigue.

(ii) Progressive degeneration of skeletal muscles mostly due to genetic disorder.

Which of the following belong(s) to myasthenia gravis?

- (1) Only (ii)
- (2) Both (i) and (ii)
- (3) Only (i)
- (4) None of these

37. Assertion : Ball and socket joints are the most mobile joint.

Reason : Synovial fluid is present here.

- (1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) If Assertion is true but Reason is false
- (4) If both Assertion and Reason are false

38. Among females, most common cause of osteoporosis.

- (1) Decrease levels of oestrogen
- (2) Decrease levels of progesterone
- (3) Increase levels of progesterone
- (4) Increase levels of oestrogen

39. Red muscle fibre differs from white muscle fibre as former

- (1) Has plenty of mitochondria
- (2) Has plenty of sarcoplasmic reticulum
- (3) Performs only anaerobic respiration
- (4) Has less quantity of myoglobin

40. Which of the following is not the part of axial skeleton?

- (1) Pectoral girdle
- (2) Facial bones
- (3) Ear ossicles
- (4) Sternum

41. Match the column

Column-A		Column-B	
(A)	True ribs	(i)	3 pairs
(B)	False ribs	(ii)	2 pairs
(C)	Floating ribs	(iii)	7 pairs

- (1) A-i; B-ii; C-iii
- (2) A-iii; B-i; C-ii
- (3) A-iii; B-ii; C-i
- (4) A-ii; B-i; C-iii

42. Select the correct option about the nature and composition of cartilage.

- (1) Very hard due to Ca^{+2} salts
- (2) Very hard due to chondroitin salts
- (3) Pliable due to calcium salts
- (4) Pliable due to chondroitin salts

43. Assertion : Mechanism of muscle contraction is explained by sliding filament theory.

Reason : Contraction of muscle fibre takes place by sliding of thick filament over thin filament.

- (1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) If Assertion is true but Reason is false
- (4) If both Assertion and Reason are false

44. The shoulder blade is a large triangular bone situated in the dorsal part of the thorax between the 2nd and the 7th ribs. It is called

- (1) Clavicle
- (2) Ilium
- (3) Scapula
- (4) Carpal

45. Acetabulum occurs in

- (1) Cranium
- (2) Pectoral girdle
- (3) Pelvic girdle
- (4) Vertebrae

46. Select incorrect statement.

- (1) Pectoral and pelvic girdle bones help in the articulation of the upper and the lower limbs respectively with the axial skeleton.
- (2) Scapula is a large triangular flat bone situated in the ventral part of the thorax.
- (3) Each clavicle is a long slender bone with two curvatures.
- (4) Clavicle is commonly called the collar bone.

47. Match the column

Column-A		Column-B	
(I)	Tarsals	(A)	14
(II)	Metacarpals	(B)	1
(III)	Phalanges	(C)	7
(IV)	Tibia	(D)	5

- (1) I-A, II-B, III-D, IV-C
- (2) I-C, II-D, III-B, IV-A
- (3) I-B, II-A, III-D, IV-C
- (4) I-C, II-D, III-A, IV-B

48. A cup shaped bone called patella cover the knee _____ (knee cap).

- (1) posteriorly
- (2) dorsally
- (3) anteriorly
- (4) ventrally

49. Which of the following statement about human vertebral column is false?

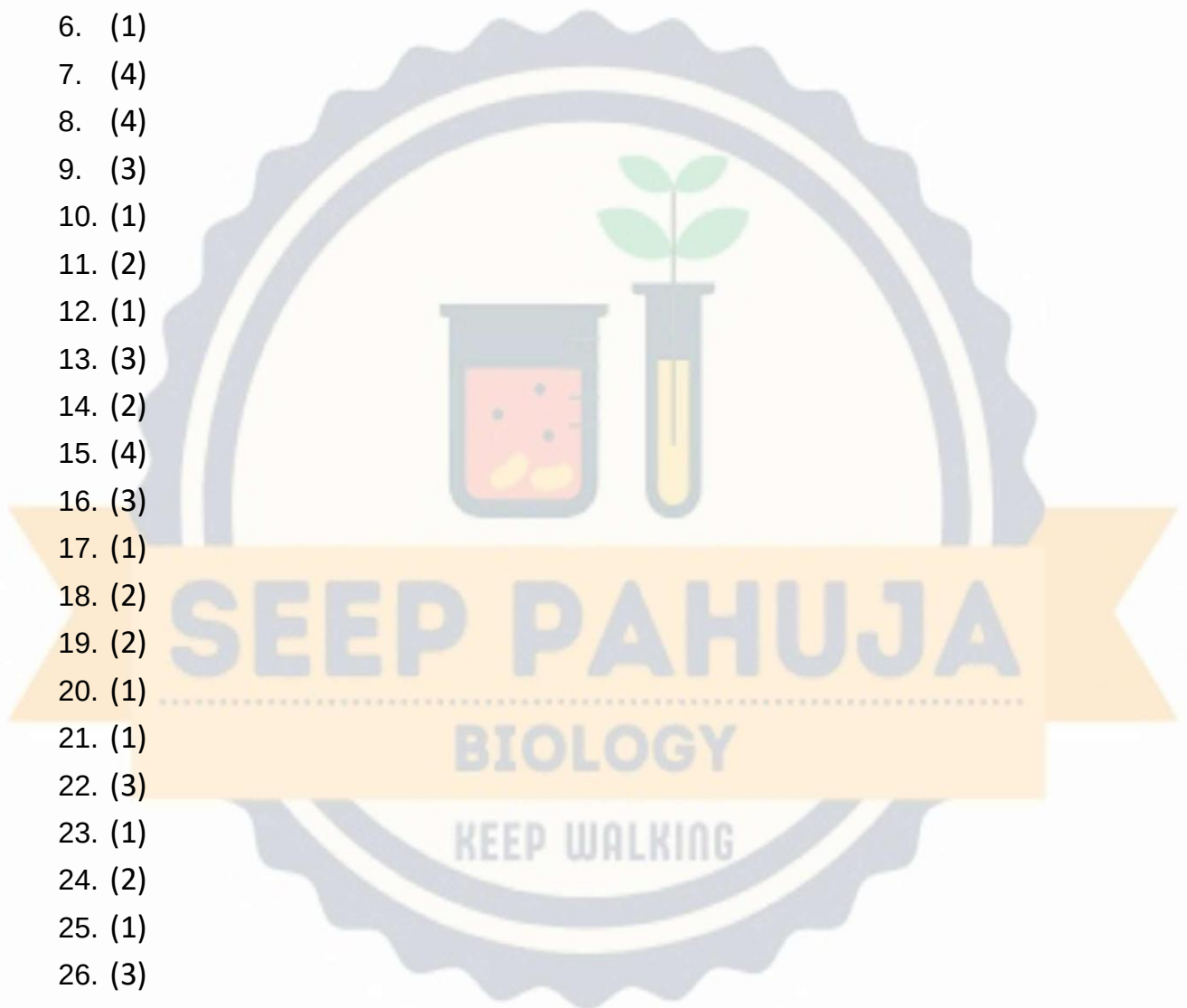
- (1) Vertebral column consists of 26 vertebrae.
- (2) It is ventrally placed.
- (3) It extends from the base of skull and constitutes the main framework of the trunk.
- (4) Neural canal in vertebra is the passage for spinal cord.

50. Human adult vertebral formula is

- (1) C4 T8 L4 S8 C8
- (2) C7 T8 L5 S6 C7
- (3) C7 T12 L2 S1 C2
- (4) C7 T12 L5 S1 C1

Answer Key

1. (1)
2. (1)
3. (3)
4. (1)
5. (1)
6. (1)
7. (4)
8. (4)
9. (3)
10. (1)
11. (2)
12. (1)
13. (3)
14. (2)
15. (4)
16. (3)
17. (1)
18. (2)
19. (2)
20. (1)
21. (1)
22. (3)
23. (1)
24. (2)
25. (1)
26. (3)
27. (2)
28. (4)
29. (4)
30. (4)
31. (2)
32. (4)
33. (1)
34. (3)



35. (4)

36. (3)

37. (2)

38. (1)

39. (1)

40. (1)

41. (2)

42. (4)

43. (3)

44. (3)

45. (3)

46. (2)

47. (4)

48. (4)

49. (2)

50. (4)

